

Route δ to the Sharp Quantitative Faber–Krahn Inequality: Step-by-Step Status, Remaining Obstructions, and Candidate Fixes

May 14, 2026

Abstract

This digest summarises the current state of Plan 2 / Route δ , which proves the sharp quantitative Faber–Krahn inequality $\delta_{\text{FK}}(\Omega) \geq c(n)\mathcal{A}(\Omega)^2$ via Saint–Venant stability and a Kohler–Jobin transfer. Each step of the proof chain is condensed to its essential idea, its input, and its output, with **closed** or **OPEN** tags. The remaining genuinely open question collapses to a single integrated homothetic-trace estimate on isoperimetrically bad levels. We list six candidate routes for closing it, with explicit critique of why each does or does not appear to work.

1 The target and the reduction

Target (Faber–Krahn). For every open $\Omega \subset \mathbb{R}^n$ of finite measure,

$$\left(\frac{|\Omega|}{|B_1|}\right)^{2/n} (\lambda_1(\Omega) - \lambda_1(B^*)) \geq c_{\text{FK}}(n) \mathcal{A}(\Omega)^2, \quad B^* \text{ ball with } |B^*| = |\Omega|.$$

The exponent 2 is sharp; equality is approached by smooth volume-preserving perturbations of the ball.

Reduction to Saint–Venant. Kohler–Jobin transfers sharp Saint–Venant deficit stability to sharp Faber–Krahn: if $\delta_T(\Omega) := E(\Omega) - E(B^*)$ controls $\mathcal{A}(\Omega)^2$ at rate δ_T , then δ_{FK} also controls $\mathcal{A}(\Omega)^2$ at rate δ_{FK} . **closed**. Reference: `Final/KohlerJobinTransfer.tex` (Plan 1, imported).

Bounded reduction. The bounded sharp Saint–Venant inequality on $\Omega \subset B_R$ with $|\Omega| = \omega_n$ implies the global one via a Cicalese–Leonardi / BDV selection. **closed**. Reference: `Final/BoundedReduction.tex`.

The bounded target.

$$\mathcal{A}(\Omega)^2 \leq C(n, R) \delta_T(\Omega).$$

This is what Route δ aims to prove. Throughout the rest of this digest, $|\Omega| = \omega_n$, $\Omega \subset B_R$, and $u = u_\Omega$ denotes the variational torsion function. Write

$$E_\rho := \{u > t(\rho)\}, \quad |E_\rho| = \omega_n \rho^n, \quad F_\rho := \rho^{-1} E_\rho \in \mathcal{X},$$

where $\mathcal{X}$ is the L^1 -modulo-translations metric space. Set $w(\rho) := -t'(\rho)$, $d\mu := w d\rho$, and $\rho_\delta := 1 - \kappa\sqrt{\delta_T}$.

2 The eight-step chain

Step	Block	Status
1. Exact level-set deficit identity	LevelSetIdentity	closed
2. Metric first variation in $\mathcal{X}$	LimsupDeformation (new)	closed
3. Same-centre Fusco–Julin package	SameCentreFJ (new)	closed
4. Trimmed velocity / good levels	TrimmedVelocityRepair	closed
4'. Unconditional velocity (all levels)	UnconditionalVelocity (new)	closed
5. Integrated action $D_H + D_I \Rightarrow d_{\mathcal{X}}^2 + \dot{F} ^2$	WeightedMetricTrace	OPEN [†]
6. Uniform window trace	WeightedMetricTrace	OPEN [†]
6'. Good Borel endpoint $\hat{\rho}$ near ρ_δ	GoodEndpointTrace (new)	OPEN [†]
7. Boundary-layer transfer from $E_{\hat{\rho}}$ to Ω	BoundaryLayerTransfer	closed
8. Bounded $\Rightarrow$ global $\Rightarrow$ FK	GlobalAssembly	closed

[†] Steps 5–6 are open only modulo a *single* integrated bound on bad levels; see Section 4.

3 Step-by-step summary

Step 1. Level-set deficit identity

Statement.

$$\frac{2\delta_T(\Omega)}{|\Omega|} = \frac{1}{|\Omega|} \int_0^{\|u\|_\infty} \frac{D_H(t) + D_I(t)}{n^2 \omega_n^{2/n} m(t)^{1-2/n}} dt,$$

where $D_H(t) = \alpha(t)\beta(t) - P(E_t)^2$ is the Cauchy–Schwarz defect of $|\nabla u|$ on $\partial^* E_t$, and $D_I(t) = P(E_t)^2 - P(B(E_t))^2$ is the squared isoperimetric defect.

Output. $\int_{\rho_*}^{\rho_\delta} (D_H + D_I) d\mu \leq C(n, \rho_*) \delta_T$.

Proof tools. BV coarea on reduced boundaries, Lipschitz truncation for the flux identity $\int_{\partial^* E_t} |\nabla u| d\mathcal{H}^{n-1} = |E_t|$, corrected $I'' = -1/\alpha$ for the rearranged primitive, Pólya–Szegő endpoint convexity.

Step 2. Metric first variation (Deliverable A)

Statement. For a.e. ρ and every bounded Borel centre field z_ρ ,

$$|\dot{F}_\rho|_{\mathcal{X}} \leq \rho^{-n} \inf_{a \in \mathbb{R}^n} \int_{\partial^* E_\rho} |V_\rho - H_{z_\rho, \rho} - a \cdot \nu_\rho| d\mathcal{H}^{n-1},$$

with $V_\rho = w/|\nabla u|$, $H_{z, \rho}(x) = (x - z) \cdot \nu_\rho(x)/\rho$.

Proof idea. Direct coarea on the symmetric difference $E_{\rho+h} \Delta T_h(E_\rho)$, where T_h is the affine pull-back combining homothety $y \mapsto z + s^{-1}(y - z)$ and a chosen translation hb . No Reshetnyak or BV lower-semicontinuity.

Status. **closed.** Replaces the conditional upper-gradient recovery hypothesis of the older MetricFramework.

Step 3. Same-centre Fusco–Julin (Deliverables B+C)

Statement. For every $E \subset K \subset \mathbb{R}^n$ of finite perimeter with $|E| = \omega_n$ and $\mathcal{D}(E) \leq \eta_0(n, K)$, there exists a single centre $z_E \in K$ with

$$\alpha_{FJ}(E, z_E)^2 + b_{FJ}(E, z_E)^2 \leq C_{\text{scFJ}}(n) \mathcal{D}(E).$$

Moreover the centre can be chosen *Borel measurable* in ρ along the torsion family.

Proof idea. Joint functional $\Phi_E(z) := \alpha_{FJ}^2 + b_{FJ}^2$ on the compact admissible K ; LSC + compactness give attainment; the centres-close estimate $|z_F - z_{FJ}| \leq C\mathcal{D}^{1/2}$ (via Fuglede / Cicalese–Leonardi nearly-spherical) gives the bound; Castaing–Valadier provides Borel selection.

Status. **closed**. Replaces the conditional theorems of FJCenterBridge.

Step 4. Velocity defect

Statement (unconditional, Theorem 2.2 of UnconditionalVelocity). For every a.e. finite-perimeter level,

$$\left(\int_{\partial^* E_\rho} |V_\rho - \bar{V}_\rho| d\mathcal{H}^{n-1} \right)^2 \leq \left(\frac{P(B_\rho)}{P(E_\rho)} \right)^2 D_H(t(\rho)) \leq D_H(t(\rho)).$$

Proof idea. The weighted variance identity $\int (V - \bar{V})^2 / V d\mathcal{H}^{n-1} = P^* D_H / P^2$ (closed form from $w\alpha = P^*$ and $\beta = |E_\rho|$); Cauchy–Schwarz $(\int |g|)^2 \leq (\int g^2 / V)(\int V)$.

Status. **closed**.

Step 5. Integrated action and kinetic energy

Targets.

$$(5a) \quad \int_{\rho_*}^{\rho_\delta} (d\mathcal{X}(F_\rho, B_1)^2 + |\dot{F}_\rho|_{\mathcal{X}}^2) d\mu \leq C \delta_T,$$

$$(5b) \quad \int_{\rho_*}^{\rho_\delta} |\dot{F}_\rho|_{\mathcal{X}}^2 d\rho \leq C \delta_T \quad (\text{Lebesgue form}).$$

Status. On $G_I = \{D_I \leq \eta_I\}$, the chain [Step 2]+[Step 3]+[Step 4] gives the pointwise $|\dot{F}_\rho|^2 \leq C(D_H + D_I)$, hence **closed** on G_I . Integrated against $d\mu$, the level-set identity closes the G_I piece unconditionally.

The *bad-isoperimetric* piece, $\int_{B_I} |\dot{F}|^2 d\mu$, is currently an explicit hypothesis ((eq:bad-I-action) of WeightedMetricTrace). **OPEN**.

Step 6. Trace at the deficit endpoint

Statement. $\sup_{\rho \in J^*} d\mathcal{X}(F_\rho, B_1)^2 \leq C\delta_T$ on the order-one window $J^* = [\rho_\delta - L^*, \rho_\delta]$.

Proof idea. Markov on (5a) inside J^* gives a single $\rho_0 \in J^*$ with $d^2 \leq K_1\delta_T$. Kinetic Cauchy–Schwarz with (5b) transports this to all of J^* .

Status. **closed** conditional on (5b), which in turn depends on Step 5’s bad-level hypotheses.

Variant (Deliverable D). Replace “trace at ρ_δ ” by “trace at a Borel good level $\hat{\rho} \in [\rho_\delta - C_1\delta_T, \rho_\delta] \cap G_I \cap G_\tau$ ” via the bad-set Lebesgue bounds $\mathcal{L}^1(B_\tau) \leq C\delta_T$, $\mathcal{L}^1(B_I \cap G_\tau) \leq C\delta_T$. This cleans the downstream interface but does *not* eliminate the conditional hypotheses of Step 5.

Step 7. Boundary-layer transfer

Statement. $\mathcal{A}(\Omega) \leq \mathcal{A}(E_{\hat{\rho}}) + \frac{2}{\omega_n} |\Omega \setminus E_{\hat{\rho}}|$, and $|\Omega \setminus E_{\hat{\rho}}| \leq \omega_n n(\kappa\sqrt{\delta_T} + C_1\delta_T) = O(\sqrt{\delta_T})$.

Squaring: $\mathcal{A}(\Omega)^2 \leq C\delta_T + O(\delta_T^{3/2})$; sharp exponent preserved. **closed**.

Step 8. Bounded $\Rightarrow$ FK

Statement. Combine Step 7 with the bounded reduction (BDV / Cicalese–Leonardi selection imported from Plan 1) and Kohler–Jobin to obtain the global FK at the sharp exponent 2. **closed**.

4 The single remaining obstruction

The genuinely open question. The whole chain reduces to: *produce a bounded Borel centre field $\rho \mapsto z_\rho \in \mathbb{R}^n$ such that*

$$\int_{\rho_*}^{\rho_\delta} \left(\int_{\partial^* E_\rho} |H_{z_\rho, \rho} - \bar{V}_\rho| d\mathcal{H}^{n-1} \right)^2 d\mu(\rho) \leq C(n, R, \rho_*) \delta_T(\Omega). \quad (1)$$

Why this is enough. Combining Steps 2, 4 (unconditional), and (1) via the triangle inequality $|V - H_z| \leq |V - \bar{V}| + |H_z - \bar{V}|$ and squaring gives the full integrated kinetic bound $\int |\dot{F}_\rho|^2 d\mu \leq C\delta_T$ on *all* levels. In particular both bad-level kinetic hypotheses of `WeightedMetricTrace` (`BAD-I-ACTION`, `BAD-TAU-ACTION`) are discharged.

Why this is hard. On the Fusco–Julin good set G_I , the same-centre theorem (Step 3) yields $(\int |H_{z_\rho, \rho} - \bar{V}_\rho|)^2 \leq C\mathcal{D}(E_\rho) \leq C'\mathcal{I}(\rho)$ pointwise, so (1) restricted to G_I follows from the level-set identity. On the bad set $B_I = \{\mathcal{I} > \eta_I\}$ the FJ inequality still holds globally as $b_{FJ}(E, z_{FJ})^2 \leq C\mathcal{D}(E)$, but *the Fraenkel and FJ centres no longer coincide*: a centre z minimising the joint $\alpha^2 + b^2$ may not exist, and the radial-trace identity

$$\int |H_z - \bar{V}| \leq C[|E\Delta B(z)| + \int |\nu - e_z|^2]$$

loses on one side or the other at any single z .

Quantitative obstruction. Pointwise the perimeter $P(E_\rho)$ can be unboundedly larger than $P(B_\rho)$ on bad levels (a torsion superlevel set may be very rough), and the homothetic integrand $|H_z - \bar{V}|$ has uncontrolled L^2 norm on $\partial^* E_\rho$. The L^1 norm is controlled by the asymmetry and the normal oscillation but not in a form that integrates to $C\delta_T$ after squaring.

5 Candidate routes for closing (1)

5.1 Route I: centroid + space–time Π -identity

Idea. Set $z_\rho := \bar{z}_\rho = \text{centroid of } E_\rho$. The divergence identity

$$\int_{\partial^* E_\rho} |\nu_{E_\rho} - e_{\bar{z}_\rho}|^2 d\mathcal{H}^{n-1} = 2(P(E_\rho) - P(B_\rho)) + 2\Pi(E_\rho, \bar{z}_\rho)$$

of `SpaceTimeIdentity` converts the normal oscillation into a quadratic form Π , and the pointwise slice-rearrangement inequality $|E_\rho \Delta B_\rho(\bar{z}_\rho)|^2 \leq C\Pi$ controls the Fraenkel term by Π too.

Why it might work. If the conditional integrated estimate $\int \Pi d\mu \leq C\delta_T$ (“ Π -int”) can be proved, then $\int \int |\nu - e_z|^2 d\mathcal{H}^{n-1} d\mu \leq C\delta_T$ globally (good and bad levels both), and Cauchy–Schwarz then bounds (1) at the centroid centre.

Why it might not. Π -int is currently an unproved structural conjecture (the abstract Talenti shortcut was invalidated in the May 14 audit). More seriously, even granted Π -int, the *squared* integral $\int (\int |\nu - e|^2 d\mathcal{H}^{n-1})^2 d\mu$ is not bounded by δ_T via naive Cauchy–Schwarz: one only obtains $\sqrt{\delta_T}$, since $(\int |\nu - e|^2)^2 \leq 4P \int |\nu - e|^2$ and $\int P(E_\rho)^2 d\mu \leq \omega_n^2$ is constant. A second deficit-aware inequality on P is needed.

Critique. The bottleneck is not really Π but the $L^2 \rightarrow L^2$ stability of $\mathcal{H}^{n-1}|_{\partial^* E_\rho}$ across μ ; Π controls the L^1 mass, and the squared form is genuinely new information. Route I would close the bound *up to a $\sqrt{\delta_T}$ exponent loss*, not at the sharp δ_T rate.

5.2 Route II: Fraenkel centre + direct radial trace

Idea. Use the Fraenkel centre $z_F(E_\rho)$, at which $|E_\rho \Delta B_\rho(z_F)|^2 \leq C_F \mathcal{I}(\rho)$ holds *globally* (Figalli–Maggi–Pratelli, no smallness). Combine with the radial trace from `FJCenterBridge`.

Why it might work. Avoids any FJ centre, so works on bad levels. The Fraenkel asymmetry is the only quantity needed for the α -piece.

Why it might not. The normal oscillation $\int |\nu - e_{z_F}|^2$ at the *Fraenkel* centre is uncontrolled on bad levels: FJ controls the oscillation at z_{FJ} , not at z_F , and the “centres-close” bound $|z_F - z_{FJ}| \leq C\mathcal{D}^{1/2}$ holds *only on good levels* (via nearly-spherical). At a bad level with comparable Fraenkel and FJ values but distinct centres, the oscillation at z_F can be of order $P(E_\rho)$, which is the L^∞ -trivial bound.

Critique. This route would require a non-trivial new “centres are close” theorem in the absence of nearly-spherical structure; this seems not to be in the literature and probably demands its own large argument.

5.3 Route III: Schwarz/Steiner symmetrisation

Idea. Replace E_ρ by its Schwarz symmetrisation $E_\rho^\#$ (a ball). Then by Pólya–Szegő, the torsion energy decreases under symmetrisation; the deficit $\delta_T(\Omega^\#) = 0$, so all level-set defects vanish for $\Omega^\#$. The difference between E_ρ and $E_\rho^\#$ is controlled by the Saint–Venant deficit.

Why it might work. Symmetrisation is the standard tool for *relating* E_ρ to a ball; in principle the deficit δ_T itself is the price of asymmetry, so a quantitative Pólya–Szegő would give a level-by-level bound.

Why it might not. Quantitative Pólya–Szegő stability (Cianchi–Esposito–Fusco–Trombetti 2011, Brock 2000) gives a bound like $\|\nabla u\|_2^2 - \|\nabla u^\#\|_2^2 \geq c \cdot \text{asymmetry}$, which involves $\|\nabla u\|$, not the level-set defect. To translate into the $\mathcal{X}$ -metric on individual levels one would still need a slice-by-slice argument, and at the slice level we are back to the same homothetic-trace question.

Critique. Symmetrisation reproves Saint–Venant deficit bounds but does not produce a centre field with control on bad levels. Useful as a sanity check but not, by itself, a closure.

5.4 Route IV: Cicalese–Leonardi / BDV selection at the level

Idea. On a bad level E_ρ , replace it by a “selection representative” $\tilde{E}_\rho$ with small deficit, then run FJ on $\tilde{E}$ and transfer back to E_ρ with a bounded error.

Why it might work. This is exactly the philosophy of the Cicalese–Leonardi quantitative isoperimetric theorem, which gave the sharp FMP exponent 2 from a selection step. Adapting it to torsion levels would automatically handle bad levels at sharp rate.

Why it might not. The Cicalese–Leonardi selection is a *compactness* argument: for sequences with $\mathcal{D} \rightarrow 0$, after translation one extracts a convergent subsequence to the ball. It does *not* provide a quantitative pointwise replacement $E_\rho \rightsquigarrow \tilde{E}_\rho$ at a single level without small deficit. Adapting the argument to bounded but not small deficit would amount to extending the regularity theory of quasi-minimal surfaces, which is a major undertaking beyond Route δ ’s scope.

Critique. Route IV is the cleanest in principle but hardest in practice: it would require new selection theorems for torsion superlevel sets, not direct application of existing FMP/FJ machinery.

5.5 Route V: optimal transport W_1 -control

Idea. Replace $|E\Delta B(z)|$ and $\int |\nu - e_z|^2$ by $W_1(\mathbf{1}_E, \mathbf{1}_{B(z)})$, the Kantorovich–Wasserstein distance. For convex sets there are quantitative Bonnesen-type inequalities relating W_1 to the isoperimetric deficit.

Why it might work. Optimal transport gives a single unified quantity controlling both “shape” and “boundary orientation” simultaneously; in principle $W_1 \leq C\sqrt{\mathcal{D}}$ covers both Fraenkel and oscillation in one inequality.

Why it might not. No quantitative $W_1(\mathbf{1}_E, \mathbf{1}_B) \leq C\mathcal{D}^{1/2}$ inequality is known for non-convex

sets at the sharp rate. Existing Wasserstein isoperimetric stability (Figalli–Indrei) gives weaker rates, and the relationship $W_1 \Leftrightarrow$ radial-trace integrand is not direct: the radial trace measures normal alignment, whereas W_1 measures bulk transport.

Critique. Aesthetically attractive but the dictionary between W_1 and the radial trace is not established at the required sharp rate.

5.6 Route VI: pointwise everywhere via algebraic identity

Idea. Look for a single closed-form identity of the type $\int (H_z - \bar{V})^2 / V d\mathcal{H}^{n-1} \leq C(n, R) (D_H + D_I)$ holding pointwise on every level, parallel to the velocity identity $\int (V - \bar{V})^2 / V = P^* D_H / P^2$. This would give the homothetic analogue of Theorem 2.2 of **UnconditionalVelocity**.

Why it might work. The velocity identity is genuinely algebraic, requiring only the flux $w\alpha = P^*$ and $\beta = |E_\rho|$. A parallel identity for H_z would use the divergence identity $\int (x - z) \cdot \nu = n|E_\rho|$ and the Reilly or Bochner formulas for torsion. Plausible in principle.

Why it might not. Concrete attempts produce closed forms involving $\int |x - z|^2 d\mathcal{H}^{n-1}$ or $\int ((x - z) \cdot \nu)^2$, which have no clean divergence representation (the corresponding vector field is not exact). D_I itself does not appear naturally in these computations; D_I is intrinsically a global “perimeter-squared minus ball-perimeter-squared” quantity, while the homothetic integrand is local. No identity is currently visible.

Critique. This is the route most analogous to the proved velocity bound (Step 4'). The reason it does not directly generalise is that V_ρ comes from the gradient (a vector field with a specific divergence), while H_z comes from the *position* x (also a vector field, but its second moments are not local on the boundary). This is a real geometric obstruction.

6 Worked attempts that fail — and what they show

We record concretely what happens when each natural Cauchy–Schwarz is applied to the homothetic integrand, so the geometric obstruction is visible.

6.1 Attempt A: weighted Cauchy–Schwarz in V_ρ

Mirror the proof of Theorem 4.1 of **UnconditionalVelocity**.

$$\left(\int |H_z - \bar{V}| d\mathcal{H}^{n-1} \right)^2 \leq \left(\int \frac{(H_z - \bar{V})^2}{V_\rho} d\mathcal{H}^{n-1} \right) \left(\int V_\rho d\mathcal{H}^{n-1} \right).$$

The second factor is $P(B_\rho) = P^*$. The first factor expands to

$$\int (H_z - \bar{V})^2 \frac{|\nabla u|}{w} d\mathcal{H}^{n-1} = \frac{1}{w} \int (H_z - \bar{V})^2 |\nabla u| d\mathcal{H}^{n-1},$$

a $|\nabla u|$ -weighted L^2 -norm of $H_z - \bar{V}$. This is finite (since H_z is bounded on $\partial^* E_\rho \subset B_R$) but not bounded by any deficit quantity: a level surface with constant $|\nabla u| = \bar{f}$ has $D_H = 0$ but generic $H_z - \bar{V} \neq 0$, so the LHS is not even nonzero where D_H is.

What this shows. V_ρ -weighting is too gradient-specific. The homothetic integrand is geometric, not gradient-geometric.

6.2 Attempt B: unweighted Cauchy–Schwarz

$$\left(\int |H_z - \bar{V}| d\mathcal{H}^{n-1} \right)^2 \leq P(E_\rho) \int (H_z - \bar{V})^2 d\mathcal{H}^{n-1}.$$

For $z = \bar{z}_\rho$ the centroid, expand $(H_z - \bar{V})^2 = H_z^2 - 2\bar{V}H_z + \bar{V}^2$ and integrate: $\int H_z = P^*$, $\bar{V}P = P^*$, so

$$\int (H_z - \bar{V})^2 d\mathcal{H}^{n-1} = \int H_z^2 d\mathcal{H}^{n-1} - \frac{P^{*2}}{P}.$$

The integral $\int H_z^2 = \rho^{-2} \int ((x - z) \cdot \nu)^2 d\mathcal{H}^{n-1}$ has no clean divergence representation: the candidate vector field $X(x) := ((x - z) \cdot \nu)(x - z)$ requires the normal ν off $\partial^* E_\rho$. Bound: $((x - z) \cdot \nu)^2 \leq |x - z|^2$, so $\int H_z^2 \leq ((R + |z|)/\rho_*)^2 P(E_\rho)/\rho^2$. Hence

$$\left(\int |H_z - \bar{V}| \right)^2 \leq C(R, \rho_*) P(E_\rho)^2.$$

The square of the perimeter is unbounded on bad levels; integrated, $\int P(E_\rho)^2 d\mu \leq \omega_n^2$ is a *constant*, not a δ_T -vanishing quantity.

What this shows. Without an additional deficit-aware bound on $\int H_z^2$, naive Cauchy–Schwarz only reaches a constant, not $O(\delta_T)$.

6.3 Attempt C: split into radial and angular contributions

Decompose $H_z - 1 = (|x - z|/\rho)(\nu \cdot e_z - 1) + (|x - z|/\rho - 1)$. The first piece has magnitude $\leq (R/\rho_*)|\nu - e_z|^2/2$, the second has magnitude $w_z := ||x - z|/\rho - 1| \leq W(R, \rho_*)$. Then

$$\int |H_z - 1| d\mathcal{H}^{n-1} \leq \frac{R + |z|}{2\rho} \int |\nu - e_z|^2 d\mathcal{H}^{n-1} + \int w_z d\mathcal{H}^{n-1}.$$

The first piece, at $z = z_{FJ}$, is $\leq CI/P^*$ globally (unconditional Fusco–Julin) — but, squared and integrated, only yields $\int (\mathcal{I}/P^*)^2 d\mu \leq \eta_I \int \mathcal{I} d\mu / P^{*2} \leq C\delta_T$ on the good set, and is unbounded on the bad set since $\mathcal{I}$ is not pointwise bounded above.

The second piece is bounded by the radial trace $\int w_z \leq (n/\rho_*)|E\Delta B_\rho(z)| + (W/2) \int |\nu - e_z|^2$, so its squared integrated form inherits the same obstruction.

What this shows. Splitting helps separate the two contributions but reduces to the same difficulty: on bad levels, $\mathcal{I}$ -pointwise control is missing.

6.4 Attempt D: choose z optimal at each level

Define $z_\rho^* := \arg \min_{z \in K} \left(\int |H_{z,\rho} - \bar{V}_\rho| d\mathcal{H}^{n-1} \right)^2$. By Castaing–Valadier this is a Borel selection (the functional is continuous in z and Borel in ρ). Then

$$\int_{\rho_*}^{\rho_\delta} \left(\int |H_{z_\rho^*, \rho} - \bar{V}_\rho| \right)^2 d\mu(\rho) = \int_{\rho_*}^{\rho_\delta} \inf_z \left(\int |H_z - \bar{V}| \right)^2 d\mu(\rho).$$

What this shows. The Borel-selection issue is not the obstruction; the obstruction is the pointwise question

$$\inf_{z \in K} \left(\int |H_{z,\rho} - \bar{V}_\rho| d\mathcal{H}^{n-1} \right)^2 \leq ?$$

for which no proved upper bound by $D_H + D_I$ is known. This is exactly $(\star)$ in the next section.

7 What kind of input *would* close (1)?

The cleanest sufficient input is a global pointwise inequality on every a.e. finite-perimeter level:

$$(\star) \quad \inf_{z \in K} \left(\int |H_{z,\rho} - \bar{V}_\rho| d\mathcal{H}^{n-1} \right)^2 \leq C(n, R, \rho_*) (D_H(t(\rho)) + D_I(t(\rho))).$$

By the level-set identity, $(\star)$ implies (1) and closes the entire route at the sharp exponent.

Equivalently, a weaker but still sufficient input is the integrated form:

$$(\star\star) \quad \int_{\rho_*}^{\rho_\delta} \inf_z \left(\int |H_{z,\rho} - \bar{V}_\rho| d\mathcal{H}^{n-1} \right)^2 d\mu(\rho) \leq C \delta_T(\Omega).$$

Routes I–VI each address $(\star\star)$ with a different choice of z_ρ and a different identification of the integrand. The crucial structural fact is that the optimum z_ρ in $(\star\star)$ should depend on the geometry of E_ρ , and the level-set identity should be *evaluated* at this z_ρ in a way that mirrors how D_H already controls the velocity defect.

8 Other consequences if the gap is closed

Closing (1) would:

- (1) Discharge the conditional Hypothesis 3.6 of `WeightedMetricTrace`;
- (2) Make the entire chain Steps 1–8 *unconditional* except for the imported BDV reduction and Kohler–Jobin (both already unconditional in Plan 1);
- (3) Yield the sharp Faber–Krahn inequality $\delta_{\text{FK}}(\Omega) \geq c(n)\mathcal{A}(\Omega)^2$ for every open $\Omega \subset \mathbb{R}^n$ of finite measure, with no smoothness, convexity, or smallness assumption beyond those already in the imports;
- (4) Provide an explicit polynomial bound on $c(n)$ in dimension (via the polynomial constants tracked in `GlobalAssembly`).

9 Concrete suggestions for next attempts

- (S1) Push Route I by proving the integrated squared-oscillation bound $\int (\int |\nu - e_{\bar{z}}|^2)^2 d\mu \leq C\delta_T$ directly, exploiting that on bad levels both factors of $\int |\nu - e_{\bar{z}}|^2$ already carry Π .
- (S2) Push Route II by proving a quantitative “centres-close” on bad levels: under what *geometric* hypothesis is $|z_F - z_{FJ}| \leq CD^{1/2}$ valid outside the nearly-spherical regime?
- (S3) Push Route IV by adapting the Cicalese–Leonardi selection to *a single level* at finite (not small) deficit; this would require a localised version of the selection principle.
- (S4) Investigate Route VI by writing the divergence identity for the candidate vector field $X(x) := (H_z(x) - \bar{V}_\rho)((x-z)/|x-z|)$ on $\partial^* E_\rho$, and computing the bulk integrand pointwise.

10 Dependency map of the route

Step 1: <code>LevelSetIdentity</code>	$\rightarrow \int (D_H + D_I) d\mu \leq C\delta_T$	closed
Step 2: <code>LimsupDeformation</code>	$\rightarrow \dot{F}_\rho \leq \rho^{-n} \inf_a \int V - H_z - a\nu d\mathcal{H}^{n-1}$	closed
Step 3: <code>SameCentreFJ</code>	$\rightarrow \alpha_{FJ}(E, z_E)^2 + b_{FJ}(E, z_E)^2 \leq CD(E)$	closed
Step 4: <code>TrimmedVelocityRepair</code>	$\rightarrow \text{good-level } (\int V - \bar{V})^2 \leq CD_H$	closed
Step 4': <code>UnconditionalVelocity</code>	$\rightarrow \text{all-level } (\int V - \bar{V})^2 \leq D_H$	closed
(1): integrated homothetic-trace	$\rightarrow \int (\int H_z - \bar{V})^2 d\mu \leq C\delta_T$	OPEN
Step 5: <code>WeightedMetricTrace</code>	$\Leftarrow$ Steps 1, 2, 4', and (1)	
Step 6: trace at endpoint	$\Leftarrow$ Step 5	
Step 7: <code>BoundaryLayerTransfer</code>	$\rightarrow \mathcal{A}(\Omega) \leq \mathcal{A}(E_{\hat{\rho}}) + (2/\omega_n) \Omega \setminus E_{\hat{\rho}} $	closed
Step 8: <code>GlobalAssembly</code>	$\rightarrow \text{Bounded SV} \Rightarrow \text{FK}$	closed

The single **OPEN** input (1) propagates downward to Steps 5 and 6. Everything else in the chain is unconditional in the current state of the proof. All inputs to (1) are local-to-a-single-level and depend only on the geometry of $E_\rho \subset B_R$ (a finite-perimeter set), not on the deficit $\delta_T(\Omega)$.

11 Cost analysis: what each candidate route requires

Route	New input needed	Rate at goal	Difficulty
I (centroid + II)	Prove $\int \Pi d\mu \leq C\delta_T$ AND $\int (\int \nu - e ^2) d\mu \leq C\delta_T$	sharp δ_T if both	medium-hard
II (Fraenkel centre)	Prove $ z_F - z_{FJ} \leq C\mathcal{D}^{1/2}$ outside the nearly-spherical regime	sharp δ_T if input proved	hard
III (symmetrisation)	Quantitative slice-wise Pólya–Szegő	weaker than δ_T	medium
IV (selection)	Adapt Cicalese–Leonardi to a single bounded-deficit level	sharp δ_T	very hard
V (optimal transport)	Sharp $W_1 \leq C\mathcal{D}^{1/2}$ for non-convex sets	sharp δ_T if input proved	very hard
VI (algebraic identity)	A closed form $\int (H_z - \bar{V})^2 / V d\mathcal{H}^{n-1} \leq C(D_H + D_I)$	sharp δ_T	hard

The "Rate at goal" column records the best achievable exponent of δ_T in $\mathcal{A}(\Omega)^2 \leq ?$ at the end of the route if the listed input is granted. Sharp δ_T corresponds to the sharp exponent 2 for $\mathcal{A}$.

12 Why this gap, and why now

The proof chain prior to the current pass had *four* conditional inputs:

- (1) the upper-gradient recovery hypothesis (closed in Deliverable A);
- (2) the same-centre Fusco–Julin theorem (closed in B);
- (3) the Borel measurable centre selection (closed in C);
- (4) the bad-level kinetic action bounds.

Three of these are now structural theorems with proofs (algebraic, quantitative-FJ, and measurable-selection respectively). The fourth is a *quantitative geometric* statement about how rough a torsion super-level set can be when its perimeter is large.

The reason the fourth is more obdurate is that the first three are *universal* (they apply to general finite-perimeter sets and torsion functions, without reference to deficit), whereas the fourth asks for a δ_T -vanishing bound on a quantity that, at any single bad level, can be arbitrarily large. The required smallness must come from a *global* balance: bad levels are rare in μ -measure, and a refined Cauchy–Schwarz must exploit this rareness. Existing inequalities are not yet sharp enough to do so at the δ_T rate after squaring.

13 Summary

The Plan 2 / Route δ proof is closed except for a single integrated homothetic-trace estimate, (1). Six candidate routes are visible; none currently closes it at the sharp δ_T rate. Routes I and VI are the most parallel to the existing closed steps (algebraic identities + level-set deficit); Route IV is the most general but requires new selection theory. The mathematical question is sharp and decoupled from Fusco–Julin; it is the unique residual gap in an otherwise complete chain.

The way forward is to combine: (a) a deficit-aware refinement of the Cauchy–Schwarz step in Route I, (b) a quantitative “centres-close” theorem outside the nearly-spherical regime for Route II, or (c) a closed-form algebraic identity for the homothetic integrand analogous to the velocity identity for Route VI. Each represents a self-contained mathematical problem that does not depend on the rest of the chain.